

Sketching Algorithms Notes

UNDERGRADUATES MODERN TCS LEARNING PATH

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Chapter 1

Counting Problems

Chapter 2

Johnson-Linderstrauss Transforms

2.1 Proof of Johnson-Linderstrauss Transforms

The following JL-Lemma is significant in high-dimensional algorithms.

Theorem 2.1.1 JL Lemma

For any $\varepsilon \in (0, 1)$ and any $X \subset \mathbb{R}^d$ for $|X| = n$ finite, there exists an embedding $f : X \rightarrow \mathbb{R}^m$ for $m = O(\varepsilon^{-2} \log n)$ such that

$$\forall x, y \in X, \quad (1 - \varepsilon)\|x - y\|_2^2 \leq \|f(x) - f(y)\|_2^2 \leq (1 + \varepsilon)\|x - y\|_2^2. \quad (2.1)$$

Note:-

The equation (1.1) tells us the ℓ_2 -norm has been preserved under such embedding f .

A simple definition can describe the "norm preservation" of embedding $f : X \rightarrow Y$

Definition 2.1.1: Distortion

Let ρ be the smallest value such that there exists some fixed value c such that

$$\forall x, y \in X, \quad c \cdot d_Y(f(x), f(y)) \leq d_X(x, y) \leq \rho c \cdot d_Y(f(x), f(y)).$$

If such ρ exists, then we say it is the *distortion* of f .

All known proofs of the JL lemma first proved the following "Distributional Johnson-Linderstrauss Lemma", which is called DJL lemma.

Theorem 2.1.2 DJL Lemma

For any $\varepsilon, \delta \in (0, 1/2)$ and integer $d > 1$, there exists a distribution $\mathcal{D}_{\varepsilon, \delta}$ over matrices $\Pi \in \mathbb{R}^{m \times d}$ for $m = O(\varepsilon^{-2} \log(1/\delta))$ such that for any fixed $z \in \mathbb{R}^d$ with $\|z\|_2 = 1$,

$$\mathbb{P}_{\Pi \sim \mathcal{D}_{\varepsilon, \delta}} \left(\left| \|\Pi z\|_2^2 - 1 \right| > \varepsilon \right) < \delta. \quad (2.2)$$

Note:-

Distribution $\mathcal{D}_{\varepsilon, \delta}$ is a matrix distribution satisfying (1.2), thus rely on ε and δ .

The following claim shows that the JL lemma is just a corollary of the DJL lemma. Proving JL lemma suffices to proof DJL lemma.

Claim 2.1.1 JL lemma is just a corollary of the DJL lemma

Acknowledge that DJL exists. We set $\delta < 1/n^2$ and pick a random Π as in DJL. For any $x \neq y \in X$, we set $z_{x,y} := (x - y)/\|x - y\|_2$. The DJL lemma implies $\mathbb{P}_{\Pi \sim \mathcal{D}_{\varepsilon, \delta}} \left(\left| \|\Pi z\|_2^2 - 1 \right| > \varepsilon \right) < \delta$, which is equivalent to $\mathbb{P} \left(\left| \|\Pi(x - y)\|_2^2 - \|x - y\|_2^2 \right| > \varepsilon \|x - y\|_2^2 \right) < \delta$. Notice that the probability there exist some $x \neq y \in X$ such that for all $\Pi \in \mathcal{D}_{\varepsilon, \delta}$ satisfy $\left| \|\Pi(x - y)\|_2^2 - \|x - y\|_2^2 \right| > \varepsilon \|x - y\|_2^2$ is at most $\binom{n}{2} \delta < 1$, because $\delta < 1/n^2$ and $|X| = n$. Thus there must exist a $\Pi^* \in \mathcal{D}_{\varepsilon, \delta}$ such that $\left| \|\Pi^*(x - y)\|_2^2 - \|x - y\|_2^2 \right| \leq \varepsilon \|x - y\|_2^2$. We let $f(x) := \Pi^* x$, which is the embedding f we want to find.

Now we begin our proof of the DJL lemma.

Proof: We proof the DJL lemma using the Hanson-Wright inequality (Tail version). We first show what is the tail version of Hanson-Wright inequality. That is corollary 2.1.1.

Corollary 2.1.1 Hanson-Wright inequality, tail version

For $\sigma_1, \dots, \sigma_n$ which is independent Rademachers ($\sigma_i = \pm 1$ with probability both $1/2$) and $A \in \mathbb{R}^{n \times n}$, for all $\lambda > 0$

$$\mathbb{P} \left(\left| \sigma^\top A \sigma - \mathbb{E} \sigma^\top A \sigma \right| > \lambda \right) \leq e^{-C\lambda^2/\|A\|_F^2} + e^{-C\lambda/\|A\|_F}.$$

Inside, the $\|\cdot\|_F$ means Frobenius Norm and $\|\cdot\|$ is the operator (spectral) norm.

We let $\mathcal{D}_{\varepsilon, \delta}$ be the distribution over matrices Π with i.i.d. entries $\Pi_{r,i} = \sigma_{r,i}/\sqrt{m}$, where the $\sigma_{r,i}$

are independent Rademachers. Define the matrix

$$B_z = \frac{1}{\sqrt{m}} \cdot \begin{bmatrix} z^\top & 0 & \cdots & 0 \\ 0 & z^\top & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & z^\top \end{bmatrix}.$$

Where $z \in \mathbb{R}^d$ can be seen as an $d \times 1$ matrix. Define the vector $\sigma \in \{-1, 1\}^{md}$ by $\sigma = (\sigma_{1,1}, \sigma_{1,2}, \dots, \sigma_{1,d}, \dots, \sigma_{m,1}, \dots)$ where $\sigma_{i,1}, \sigma_{i,2}, \dots, \sigma_{i,d}$ is the i^{th} row in Π , recall $\Pi \in \mathbb{R}^{m \times d}$. It is not hard to verify that $\Pi z = B_z \sigma$. Also notice that $\|\Pi z\|_2^2 = z^\top \Pi^\top \Pi z$, which means $\mathbb{E}[\|\Pi z\|_2^2] = 1$. Thus $\|\Pi z\|_2^2 - 1 = \|B_z \sigma\|_2^2 - 1 = \sigma^\top B_z^\top B_z \sigma - \mathbb{E}[\sigma^\top B_z^\top B_z \sigma]$.

Defining $A_z := B_z^\top B_z$, and applying Corollary 1.1.1,

$$\mathbb{P}(|\sigma^\top A_z \sigma - \mathbb{E}[\sigma^\top A_z \sigma]| > \varepsilon) \lesssim e^{-C\varepsilon^2/\|A_z\|_F^2} + e^{-C\varepsilon/\|A_z\|} \quad (2.3)$$

Thus we only need to bound $\|A_z\|_F$ and $\|A_z\|$. Remember that

$$A_z = \frac{1}{m} \begin{bmatrix} zz^\top & 0 & \cdots & 0 \\ 0 & zz^\top & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & zz^\top \end{bmatrix}$$

Where $zz^\top$ is an $d \times d$ matrix. So the squared Frobenius norm is $(1/m^2) \sum_{r=1}^m \|zz^\top\|_F^2 = (1/m^2) \sum_{r=1}^m \sum_{i,j} z_i^2 z_j^2 = (1/m) \|z\|_2^4 = 1/m$.

Since A_z is real and symmetric, we also have $\|A_z\|$ is its largest singular value, while the spectral theorem implies all its eigenvalues are real. Since A_z is block-diagonal, its eigenvalues are the eigenvalues of each block. Thus we just need to bound the largest eigenvalue of $(1/m)zz^\top$, which is very special and need some technique as the following content shows.

Note:-

Largest eigenvalue of $(1/m)zz^\top$. From rank inequality $\text{rank}(AB) \leq \text{rank}(A)$, we know that $(1/m)zz^\top$ is a rank-1 matrix. Then it has only 1 nonzero eigenvalue, and it's not difficult to find that $(1/m)zz^\top$ has a sole eigenvector with nonzero eigenvalue is z , which has corresponding eigenvalue $(1/m)\|z\|_2^2 = 1/m$.

Thus overall, equation(1.3) is bounded by

$$e^{-C\varepsilon^2/m} + e^{-C\varepsilon/m}$$

which is at most δ for choosing $m = \Omega(\varepsilon^{-1} \log(1/\delta) + \varepsilon^{-2} \log(1/\delta)) = \Omega(\varepsilon^{-2} \log(1/\delta))$, as desired.

Note:-

How to get this m ? It is equivalent to solve $e^{-C\varepsilon^2/m} + e^{-C\varepsilon/m} < \delta$ in big O corresponding ε which is very small, thus we can throw $e^{-C\varepsilon/m}$ away and solve the rest, which is trivial.

Hence we've finish the proof. ☺

2.2 Lower bound

The main question we concerned in this section is below.

Question 1: Lower bound

Whether the $m = O(\varepsilon^{-2} \log n)$ in the JL lemma is optimal?

That is, does there exist a set X of size n such that any $1 + \varepsilon$ -distortion embedding of X with the Euclidean metric into m -dimensional Euclidean space must have $m = O(\varepsilon^{-2} \log n)$?

The proof of this question goes for a long time of try and error. The first lower bound for this problem was in the original JL paper itself, which showed that for ε smaller than some fixed constant, $m = \Omega(\log n)$ is required. We present their argument here since it uses volumetric reasoning that will be useful later for the main theorem, optimal lower bound.

Proposition 2.2.1

$m = \Omega(\log n)$ is required

Proof: The proof below use the method in geometric embedding theory. Let the point set X in the original space was $X = \{0, e_1, \dots, e_{n-1}\} \subset \mathbb{R}^n$, where e_i is the i^{th} standard basis vector (1 in the i^{th} position and 0 elsewhere). Suppose $f : X \rightarrow \mathbb{R}^m$ satisfies

$$\forall x, y \in X, (1 - \varepsilon)\|x - y\|_2 \leq \|f(x) - f(y)\|_2 \leq (1 + \varepsilon)\|x - y\|_2.$$

Without loss of generality, we may assume $f(0) = 0$ (else translate the image of f by $f(0)$, which **does not** change its distortion).

Define $\tilde{e}_i := f(e_i)$. Then since $\|\tilde{e}_i - 0\|_2 = \|\tilde{e}_i - f(0)\|_2 = (1 \pm \varepsilon)\|e_i\|_2$, we have that $1 - \varepsilon \leq \|\tilde{e}_i\|_2 \leq 1 + \varepsilon$. We also have that for $i \neq j$, $\|\tilde{e}_i - \tilde{e}_j\|_2 \geq (1 - \varepsilon)\sqrt{2}$ because the two standard basis vectors' distance in original $\mathbb{R}^n$ is $\sqrt{2}$, so the radius $(1 - \varepsilon)\sqrt{2}/2$ balls around the $\tilde{e}_i$ have disjoint interior. Meanwhile, each of these balls is fully contained in a radius $(\max_i \|\tilde{e}_i\|_2) + (1 - \varepsilon)\sqrt{2}/2 \leq (1 + \varepsilon + (1 - \varepsilon)\sqrt{2}/2)$ -radius ball about the origin. Thus, for $B_d(a, r)$ denoting the ball of radius r about point

a under metric d , we have:

$$\begin{aligned}
\text{vol}(B_{\ell_2}(0, 1 + \varepsilon + (1 - \varepsilon)\sqrt{2}/2)) &\geq \text{vol}\left(\bigcup_{i=1}^{n-1} B_{\ell_2}(\tilde{e}_i, (1 - \varepsilon)\sqrt{2}/2)\right) \\
&= \sum_{i=1}^{n-1} \text{vol}(B_{\ell_2}(\tilde{e}_i, (1 - \varepsilon)\sqrt{2}/2)) \quad (\text{by disjointness of the balls}) \\
&= (n - 1) \cdot \text{vol}(B_{\ell_2}(0, (1 - \varepsilon)\sqrt{2}/2))
\end{aligned}$$

And the ball volume in $\mathbb{R}^m$ is $\pi \cdot r^m$, which implies $n - 1 \leq \left(\frac{1+\varepsilon+(1-\varepsilon)\sqrt{2}/2}{(1-\varepsilon)\sqrt{2}/2}\right)^m$, so that $m = \Omega(\log n)$ as long as $\varepsilon < 1$ is a fixed constant. $\odot$

2.2.1 Lower bound of DJL lemma

We can also ask whether the bound in DJL lemma is optimal, before showing the long proof of question 1.

Note:-

Even if so, it does not imply that the JL lemma lower bound is optimal, because in principle one could achieve improved bounds for Euclidean dimensional reduction **without using random linear maps as in the DJL lemma** (though it turns out not be the case, as we will see later in the proof of the main question/theorem).

The statement of DJL lower bound are as follows:

Theorem 2.2.1 DJL lower bound

Let $n, d > 1$ and $\varepsilon, \delta \in (0, 1/2)$ be fixed. Suppose $\mathcal{D}_{\varepsilon, \delta}$ is a distribution over $\mathbb{R}^{m \times d}$ such that for all $z \in \mathbb{R}^d$ (the unit sphere),

$$\Pr_{\Pi \sim \mathcal{D}_{\varepsilon, \delta}} \left(\left| \|\Pi z\|_2^2 - 1 \right| > \varepsilon \right) < \delta.$$

Then $m = \Omega(\min\{d, \varepsilon^{-2} \log(1/\delta)\})$.

Proof: This proof is a little bit long so we just show the proof sketch.

Proof sketch. We use the easy direction of Yao's minimax principle.

$$\begin{aligned}
\forall z \in S^{d-1} \quad \Pr_{\Pi \sim \mathcal{D}_{\varepsilon, \delta}} \left(\left| \|\Pi z\|_2^2 - 1 \right| > \varepsilon \right) &< \delta \\
\implies \Pr_{z \sim \mathcal{Z}} \Pr_{\Pi \sim \mathcal{D}_{\varepsilon, \delta}} \left(\left| \|\Pi z\|_2^2 - 1 \right| > \varepsilon \right) &< \delta \quad (\text{for any distribution } \mathcal{Z} \text{ over } S^{d-1}) \\
\implies \Pr_{\Pi \sim \mathcal{D}_{\varepsilon, \delta}} \Pr_{z \sim \mathcal{Z}} \left(\left| \|\Pi z\|_2^2 - 1 \right| > \varepsilon \right) &< \delta \\
\implies \exists \Pi \in \mathbb{R}^{m \times d} \quad \Pr_{z \sim \mathcal{Z}} \left(\left| \|\Pi z\|_2^2 - 1 \right| > \varepsilon \right) &< \delta.
\end{aligned}$$

Then this question turns to search for an appropriate “hard distribution” $\mathcal{Z}$, such Π cannot exist unless $m = \Omega(\min\{d, \varepsilon^{-2} \log(1/\delta)\})$. It turns out that the hard distribution of $\mathcal{Z}$ simply being a uniform vector over the sphere. 😊

Note:-

Yao’s minimax principle (the easy direction) states that if a randomized algorithm succeeds with high probability on every input, then there exists a deterministic algorithm that succeeds with high probability on average over some input distribution. In our case:

- The “randomized algorithm” is the random matrix $\Pi \sim \mathcal{D}_{\varepsilon, \delta}$.
- The “input” is $z \in S^{d-1}$, and we assume for each z , failure probability $< \delta$.
- Then for any input distribution $\mathcal{Z}$, the *expected* failure probability over Π is $< \delta$.
- By the averaging argument (*expectation $\leq$ maximum*), there must exist at least one fixed Π^* such that its failure probability under $\mathcal{Z}$ is also $< \delta$.

This is not a constructive proof. It does not tell us which Π^* works, but existence is sufficient for proving DJL lower bounds.

2.2.2 Optimal JL lower bound

Now, we begin our proof of optimal JL lower bounds (Question 1) which was first published in 2017 by Jelani Nelson.

Proof: First we introduce some necessary background from convex geometry.

Definition 2.2.1: Convex body

A **convex body** is a compact, convex subset of $\mathbb{R}^d$ with non-empty interior. Recall $K \subset \mathbb{R}^d$ is **convex** if for all $x, y \in K$, the straight line from x to y is fully contained in K .

It is **compact** if it is closed (contains all its limit points) and bounded (this is not the general definition of compactness, but it’s equivalent for Euclidean space $\mathbb{R}^d$).

A convex body is **symmetric** if $x \in K \iff -x \in K$. Note a symmetric convex body must thus contain 0 (look at the line between x and $-x$ for some $x \in K$, which must exist since K has non-empty interior).

A significant relation between symmetric convex body and norm space is shown below.

Claim 2.2.1 Convex bodies are 1-to-1 correspondence with normed space in $\mathbb{R}^d$

Given a norm $\|\cdot\|$ we can define a symmetric convex body $K = \{x : \|x\| \leq 1\}$. On the contrary, given a symmetric convex body K , we can define a norm $\|\cdot\|_K$ by $\|x\|_K = \sup\{t : tx \in K\}$. One can easily verify that $\|\cdot\|_K$ is a norm by examining the norm-3-principle.

Definition 2.2.2: Entropy numbers, ϵ – nets

Let (X, d_X) be a metric space and $T \subseteq X$. For $\epsilon \geq 0$, an ϵ -**net** of T is a subset $T' \subseteq T$ (**usually is a pointset**) such that for all $x \in T$, there exists $x' \in T'$ such that $d_X(x, x') \leq \epsilon$.

We define the **entropy number** $\mathcal{N}(T, d_X, \epsilon)$ as the size of the smallest ϵ -net of T under metric d_X . (**Usually the smallest pointset.**)

Given a set A and symmetric convex body K , both in $\mathbb{R}^d$, we abuse notation and also define $\mathcal{N}(A, K)$ to be the minimum number of translations of K required to cover every point in A . Note that $\mathcal{N}(T, d_{\|\cdot\|_K}, \epsilon) = \mathcal{N}(T, \epsilon K)$.

The following lemma, which plays an important role in this proof, is a well-known standard conclusion.

Proposition 2.2.2

Let B be the unit ball of some norm $\|\cdot\|$ in $\mathbb{R}^d$. Then $\mathcal{N}(B, \|\cdot\|, \epsilon) \leq (1 + 2/\epsilon)^d$. (Note that the norm itself is a kind of distance in Euclidean space.)

Proof: We define the **packing number** $\mathcal{P}(A, d, \epsilon)$ as the maximum number of radius- ϵ pairwise disjoint balls under distance d that can fit in A , centered at points in A . Then consider an optimal packing for $\mathcal{P}(B, \|\cdot\|, \epsilon/2) = N$ with centers $c_1, \dots, c_N$. The union of all such balls fits inside a ball of radius $1 + \epsilon/2$ (To see this, you just need to draw a graph.), and thus

$$\text{vol}(B_{\|\cdot\|}(0, 1 + \epsilon/2)) \geq N \cdot \text{vol}(B_{\|\cdot\|}(0, \epsilon/2)).$$

Notice these balls are in $\mathbb{R}^d$. Thus N is at most $(1 + 2/\epsilon)^d$.

Meanwhile, we claim that $\mathcal{N}(B, \|\cdot\|, \epsilon) \leq \mathcal{P}(B, \|\cdot\|, \epsilon/2) = N$. An important observation is that the balls of radius ϵ about the c_i must cover B completely. This is because if there is an $x \in B$ which is not covered, then $\forall i \in [N], B_{\|\cdot\|}(x, \epsilon/2) \cap B_{\|\cdot\|}(c_i, \epsilon/2) = \emptyset$. But thus we have added x to the center points $c_1, \dots, c_N$, which let the optimal packing number is at least $N + 1$ instead of N . This leads contradiction!

$$\text{Thus } \mathcal{N}(B, \|\cdot\|, \epsilon) \leq \mathcal{P}(B, \|\cdot\|, \epsilon/2) = N \leq (1 + 2/\epsilon)^d$$

☺

Note:-

In the most regular textbook, notation $[N]$ means the set $\{1, 2, \dots, N\}$

Next, we make a simple observation relating the additive preservation of dot products and Euclidean embeddings with low distortion.

Proposition 2.2.3

Let $X \subset S^{d-1}$ be such that it contains 0 and $f : X \rightarrow \mathbb{R}^m$ satisfies $f(0) = 0$ and low distortion

$$\forall x, y \in X, \quad (1 - \varepsilon)\|x - y\|_2^2 \leq \|f(x) - f(y)\|_2^2 \leq (1 + \varepsilon)\|x - y\|_2^2. \quad (2.4)$$

Then

$$\forall x, y \in X \setminus \{0\}, \quad |\langle f(x), f(y) \rangle - \langle x, y \rangle| \leq 3\varepsilon. \quad (2.5)$$

Proof: By linearity, we have

$$\|x - y\|_2^2 = \|x\|_2^2 + \|y\|_2^2 - 2\langle x, y \rangle = 2 - 2\langle x, y \rangle,$$

and according to the condition that $X \subset S^{d-1}$, where S^{d-1} is the unit ball in $\mathbb{R}^d$ and the embedding f preserves distortion,

$$\|f(x) - f(y)\|_2^2 = \|f(x)\|_2^2 + \|f(y)\|_2^2 - 2\langle f(x), f(y) \rangle = 1 \pm \varepsilon + 1 \pm \varepsilon - 2\langle f(x), f(y) \rangle.$$

Subtracting and rearranging, then use the distortion condition, we have

$$|\langle f(x), f(y) \rangle - \langle x, y \rangle| = \frac{1}{2}|2 + 2\varepsilon - 2 + \|f(x) - f(y)\|_2^2 - \|x - y\|_2^2| \leq \frac{1}{2}[2\varepsilon + \varepsilon\|x - y\|_2^2] \leq 3\varepsilon.$$

The above inequality holds by the triangle inequality since $\|x - y\|_2 \leq \|x\|_2 + \|y\|_2 = 2$. $\odot$

We now present the formal proof of the lower bound, which is a **compression argument**. Let $\mathcal{X}$ be a collection of n -point sequences in $\mathbb{R}^d$. We want to show that if for each $X \in \mathcal{X}$ there exists an embedding $f_X : X \rightarrow \mathbb{R}^m$ satisfying Equation (1.4) and (1.5) for $m \ll \varepsilon^{-2} \log n$ with error ε , then there must be an **injection** $g : \mathcal{X} \rightarrow \{0, 1\}^S$ for some $S < \log_2 |\mathcal{X}|$. This will finish our proof because it's a contradiction. Thus there must be an $X' \in \mathcal{X}$ such that no such f_X exists. Again, by proposition 1.2.3, there must thus be no embedding with distortion at most $1 + \varepsilon/3$ into such small m , since it doesn't fix $X' \in \mathcal{X}$. The last step of this proof is applying the argument to $\varepsilon' = 3\varepsilon$, thus we will not state this last step below.

Define $k = \lfloor 1/(100\varepsilon^2) \rfloor$ and $y_{S_q} = (1/\sqrt{k}) \sum_{i \in S_q} e_i$. We let $\mathcal{X}$ consist of all point sequences of the form $X = (0, e_1, e_2, \dots, e_d, y_{S_1}, \dots, y_{S_{n-d-1}})$ where each S_j are subsets of $[d]$ of size k . Thus $|\mathcal{X}| = \binom{d}{k}^{n-d-1}$, and it's easy to see that for each $0 \neq x \in X$, $\|x\|_2 = 1$.

Note:-

Notice that $\langle e_i, y_{S_j} \rangle$ is either 0 (if $i \notin S_j$) or 10ε (if $i \in S_j$).

Thus knowing $\langle e_i, y_{S_j} \rangle$ for each i, j is enough to determine (decoding characteristic) X completely. We will show that if for all $X \in \mathcal{X}$, the desired f_X which embedding into dimension m and preserve these dot products up to error $\pm \varepsilon$ exists, then there exists an injection $g: \mathcal{X} \rightarrow \{0, 1\}^{O(nm)}$, and thus we must have $nm = \Omega(\log_2 |\mathcal{X}|) = \Omega(nk \log(d/k))$. Thus $m = \Omega(k \log(d/k))$. The following assumption is non-trivial.

Note:-

Firstly, we assume $d = n/\log(1/\varepsilon)$ and $1/\varepsilon^2 < n^{.99}$.

Thus we have $m = \Omega(\varepsilon^{-2} \log n)$, as desired. We now simply must show how to define such g , as the argument below does. After show the construction of g , we will remove the assumption above.

We would like to set $g(X)$ to be the concatenation of entries of the sequence of vectors $f_X(0), \dots, f_X(e_d), f_X(y_{S_1}), \dots, f_X(y_{S_{n-d-1}})$, which would be nm numbers since $f_X(e_d)$ is m -dimensional. Then since f_X preserves dot products up to additive $\pm \varepsilon$, we can know whether $i \in S_j$ based on whether $\langle f_X(e_i), f_X(y_{S_j}) \rangle$ is either at most ε or at least 9ε . Unfortunately, this does not work, since then g maps X to nm real numbers, but we want a map into $O(nm)$ bits. We now show how to get around this in progressively 3 better ways, under the analysis of γ -nets. Starts from here, we refer to f_X as simply f .

Example 2.2.1 (Construct g : First attempt)

Our first approach is to round each entry of $f(x)$ to the nearest integer multiple of γ for some small $\gamma > 0$. Since $0 \neq x \in X, \|x\|_2 = 1$, by distortion, $f(x)$ for $x \neq 0$ has norm that is $1 \pm \varepsilon$, thus each entry of $f(x)$ must be in the interval $[-1 - \varepsilon, 1 + \varepsilon]$. Thus there are at most $\lceil (2 + 2\varepsilon)/\gamma \rceil = O(1/\gamma)$ possible multiples of γ that each entry could be rounded to. Thus, after rounding, the sequence $f(0), f(e_1), \dots, f(e_d), f(y_{S_1}), \dots, f(y_{S_{n-d-1}})$ consumes at most $O(nm \log(1/\gamma))$ **bits** to specify. Meanwhile, the dot products after rounding are

$$\sum_{i=1}^m (f(e_i) \pm \gamma)(f(y_S) \pm \gamma) = \langle f(e_i), f(y_S) \rangle \pm \gamma \|f(e_i)\|_1 \pm \gamma \|f(y_S)\|_1 \pm m\gamma^2.$$

By Cauchy-Schwarz, $\|f(e_i)\|_1 \leq \sqrt{m} \|f(e_i)\|_2 \leq (1 + O(\varepsilon))\sqrt{m}$, and it's similarly for $\|f(y_S)\|_1$. Thus for $\gamma = \Theta(\varepsilon/\sqrt{m})$, the total additive error in computing $\langle f(e_i), f(y_S) \rangle$ is at most ε . Recalling $\langle f(e_i), f(y_S) \rangle$ approximates $\langle e_i, y_S \rangle$ up to additive $\pm \varepsilon$, we thus know $\langle e_i, y_S \rangle$ up to additive $\pm 2\varepsilon$. From the note previously, we can thus declare $i \in S$ if our estimated dot product is at least 8ε , and $i \notin S$ if it is at most 2ε . Our g then maps into $O(nm \log(1/\gamma)) = O(nm \log(m/\varepsilon))$ bits. Since this quantity must be at least $\Omega(n\varepsilon^{-2} \log n)$, we have

$$m = \Omega\left(\varepsilon^{-2} \cdot \frac{\log n}{\log(1/\varepsilon) + \log \log n}\right).$$

Example 2.2.2 (Construct g : Second attempt)

In the first attempt we picked a γ -net B' of $B_{\ell_2}(0, 1 + \varepsilon)$ under the ℓ_∞ norm and rounded each $f(x)$ to the closest element in B' . What if we instead picked B'' to be a γ -net under ℓ_2 instead of ℓ_∞ ? Let $\widetilde{f(x)}$ be the closest point to $f(x)$ in B'' (under ℓ_2). Then

$$\begin{aligned}\langle \widetilde{f(x)}, \widetilde{f(y)} \rangle &= \langle f(x) + (\widetilde{f(x)} - f(x)), f(y) + (\widetilde{f(y)} - f(y)) \rangle \\ &= \langle f(x), f(y) \rangle + \langle f(x), \widetilde{f(y)} - f(y) \rangle + \langle \widetilde{f(x)} - f(x), f(y) \rangle + \langle \widetilde{f(x)} - f(x), \widetilde{f(y)} - f(y) \rangle \\ &= \langle f(x), f(y) \rangle + O(\gamma)\end{aligned}$$

with the last inequality holding by Cauchy-Schwarz and since $\|\widetilde{f(y)} - f(y)\|_2, \|\widetilde{f(x)} - f(x)\|_2 \leq \gamma$. Thus if we let $\gamma = c\varepsilon$ for a sufficiently small constant c , we have $\langle \widetilde{f(x)}, \widetilde{f(y)} \rangle = \langle f(x), f(y) \rangle \pm \varepsilon$. Again, we can declare $i \in S$ if $\langle \widetilde{f(x)}, \widetilde{f(y)} \rangle$ is at least 8ε , and $i \notin S$ if it is at most 2ε .

By proposition 1.2.4, B'' has size at most $O((1/\varepsilon)^m)$, and thus the number of bits to specify $(\widetilde{f(x)})_{x \in X}$ is $O(nm \log(1/\varepsilon))$, which must be $\Omega(\log |X|) = \Omega(n\varepsilon^{-2} \log n)$. Thus

$$m = \Omega\left(\varepsilon^{-2} \cdot \frac{\log n}{\log(1/\varepsilon)}\right).$$

This recovers the lower bound of [Alon03], but with a totally different proof.

The second attempt shows an important observation: it was enough to know $\langle \widetilde{f(e_i)}, \widetilde{f(y_{S_j})} \rangle$ for each $i \in [d], j \in [n - d - 1]$ to recover $X \in X$.

Claim 2.2.2 Construct g : Final attempt

we now remove the $\log(1/\varepsilon)$ in the denominator in the previous attempt. From the observation above, we know that if we define $v_j \in \mathbb{R}^d$ by $v_j := \langle \widetilde{f(e_i)}, \widetilde{f(y_{S_j})} \rangle$, then it is enough to know all the v_j . Moreover, it is even enough to know a ε -net set of $\tilde{v}_j$ such that $\|\tilde{v}_j - v_j\|_\infty \leq \varepsilon$ for each j . That is, we only need to know roundings of the v_j in an ε -net under the ℓ_∞ metric, since then we can distinguish whether $i \in S_j$ by whether $(\tilde{v}_j)_i$ is at most 3ε or at least 7ε , basis on the 8ε versus 2ε -distinguish.

If we define a matrix $A \in \mathbb{R}^{d \times m}$ to have its i th row be $\widetilde{f(e_i)}$, then $v_j = A \widetilde{f(y_{S_j})}$. Thus the v_j **live in the subspace E which is the column space of A** , and since A has at most m columns, we have $\dim(E) \leq m$. Also by corresponding previous analysis, since $\langle \widetilde{f(e_i)}, \widetilde{f(y_{S_j})} \rangle = \langle f(e_i), f(y_{S_j}) \rangle \pm \varepsilon = \langle e_i, y_{S_j} \rangle \pm 2\varepsilon$, we know that $\|v_j\|_\infty \leq 12\varepsilon$ for each j . Thus if we define $K := E \cap B_{\ell_\infty^d}(0, 12\varepsilon)$, then $v_j \in K$ for each j . Now the beauty comes: K is a **symmetric convex body**, so it defines a norm! Thus if we let K' be a $1/12^{th}$ -net of K in the K -norm, it implies that for any $v \in K$ there exists $\tilde{v} \in K'$ such that $\|v - \tilde{v}\|_K \leq \varepsilon$.

This will be our definition of the $\tilde{v}_j$: simply the rounding of the v_j to the closest points (under the K -norm) in K' . The size of K' is $O(1 + 2/(1/12))^m = O(1)^m$ since K is m -dimensional (proposition

1.2.3), and thus $\tilde{v}_j$ can be specified in $O(m)$ bits. Thus we all $\tilde{v}_j$ for every $j \in [n - d - 1]$ can be specified in $O(nm)$ bits combined! Thus we complete our construction path!

Note:-

There is one slight catch: recall in compression arguments we typically show that the “compression” is an injection by showing that it is possible to invert. But for the decompressor to invert, they will need to know which body K we are talking about. For that, they need to know the matrix A , which depends on $f = f_X$ and thus depends on X . We accomplish this by simply writing down A explicitly, row-by-row; each $\widetilde{f(e_i)}$ takes $O(m \log(1/\varepsilon))$ bits, so in total this takes $O(dm \log(1/\varepsilon)) = O(nm)$ bits since we have assumed that $d = n/\log(1/\varepsilon)$.

Now came the last part of our proof.

Removing the assumption $d = n/\log(1/\varepsilon)$: Above we showed how to obtain the optimal lower bound when $d = n/\log(1/\varepsilon)$. What about for other d ? Showing a hard set with larger d exists is easy: we just take the hard point set in dimension $n/\log(1/\varepsilon)$ then set all the other vectors with zeroes to make them dimension $d = n/\log(1/\varepsilon)$.

What about for smaller d ? We suppose X is a hard point set in dimension $n/\log(1/\varepsilon)$, that is, any embedding with distortion $(1 + \varepsilon)^2 = 1 + O(\varepsilon)$ needs target dimension $\geq c\varepsilon^{-2} \log n$. Consider the set of points $f(X)$ in dimension $d' = O(\varepsilon^{-2} \log n)$ where $f: X \rightarrow \mathbb{R}^{d'}$ has distortion at most $1 + \varepsilon$; such f exists by the JL lemma. Then $f(X)$ must similarly be “hard” (i.e. require target dimension $\geq c\varepsilon^{-2} \log n$ to achieve distortion $1 + \varepsilon$), since otherwise if one could embed into such small dimension with low distortion via a mapping g , then $g \circ f$ would be a good embedding for X .

This is the end of hole proof. ☺

2.3 Speeding up the JL-Transforms

Our motivation for dimensionality reduction: Initially, we have some dataset $X \subset \mathbb{R}^d$ for **large** d on which we want to solve some computational problem. To make this computation easier, we map X down to some $f(X) \subset \mathbb{R}^m$ for $m \ll d$ then solve the problem on $f(X)$, which is hopefully faster than solving it on X since m is much smaller.

However, achieving m as small as possible is important being able to **compute** $f(X)$ quickly. That is, given a point $x \in X$, we would like an f that allows us to compute $f(x)$ quickly.

2.3.1 Sparse JL Transforms

One natural way to speed up JL is to make Π sparse. If Π has s non-zero entries per column, then Πx can be multiplied in time $O(s \cdot \|x\|_0)$. So the goal is to make s, m as small as possible.

Definition 2.3.1: CountSketch Matrix

Fix integers d and m . Let $h : [d] \rightarrow [m]$ be a hash function (typically chosen from a 2-wise independent family), and let $\sigma : [d] \rightarrow \{-1, +1\}$ be a random sign function (typically chosen from a 4-wise independent family). Define the CountSketch matrix $\Pi \in \mathbb{R}^{m \times d}$ by

$$\Pi_{r,i} = \begin{cases} \sigma(i), & r = h(i), \\ 0, & \text{otherwise.} \end{cases}$$

Equivalently, each column i has exactly one nonzero entry, located in row $h(i)$ with value $\sigma(i)$. Hence for any $x \in \mathbb{R}^d$,

$$(\Pi x)_r = \sum_{i: h(i)=r} \sigma(i) x_i, \quad r = 1, \dots, m.$$

We firstly consider Π as the CountSketch matrix. It was shown by [Mikkel Thorup and Yin Zhang, 2012] that this distribution provides DJL for $m \gtrsim 1/(\varepsilon^2 \delta)$. Note that the column sparsity in here is $s = 1$. The analysis is simply via Chebyshev's inequality, after doing an expectation and variance calculation.

We want to further improve the dependence of m on $1/\delta$, so we allow ourselves to increase **sparsity** s . This leads us to the **Sparse JL Transform (SJLT)**. This is a JL distribution over Π having exactly s non-zero entries per column.

As Jelani Nelson claimed, the reason for the poor dependence in m on the failure probability δ is that we use Chebyshev's inequality, which is too "trivial and easy". We will make improvements by using Hanson-Wright Inequality.

Remark: We have introduced its tail version in corollary 2.1.1. But here we use its original version, as we introduce below.

Corollary 2.3.1 Hanson–Wright inequality, original version

For $\sigma_1, \dots, \sigma_n$ independent Rademachers and $A \in \mathbb{R}^{n \times n}$, for all $p \geq 1$,

$$\|\sigma^\top A \sigma - \mathbb{E} \sigma^\top A \sigma\|_p \lesssim \sqrt{p} \|A\|_F + p \|A\|.$$

Here $\|\cdot\|$ means operator norm (spectral norm).

Note:-

- In the Hanson-Wright Inequality, the $\|\cdot\|_p$ in left hand side is the p -**moment** of the random variables $\sigma^\top A \sigma - \mathbb{E} \sigma^\top A \sigma$, while the $\|\cdot\|$ in right hand side are the norm of a matrix. They are different!
- From the probability textbook, we know that when $p \geq 1$, the p -moment of a random variable is exactly (can be seen as) a norm in the probability space due to the Minkowski Inequality.

As previously, without loss of generality we assume $z \in \mathbb{R}^d$ has $\|z\|_2 = 1$. How to Construct Π ? Our random $\Pi \in \mathbb{R}^{m \times d}$ satisfies $\Pi_{r,i} = \eta_{r,i} \sigma_{r,i} / \sqrt{s}$ for some integer $1 \leq s \leq m$. The $\sigma_{r,i}$ are independent Rademachers, in which the $\eta_{r,i}$ are Bernoulli random variables satisfying:

- For all r, i , $\mathbb{E} \eta_{r,i} = s/m$.
- For any i , $\sum_{r=1}^m \eta_{r,i} = s$. That is, each column of Π has *exactly* s non-zero entries.
- The $\eta_{r,i}$ are **negatively correlated**. That is, for any subset $S \subseteq [m] \times [n]$, we have

$$\mathbb{E} \prod_{(r,i) \in S} \eta_{r,i} \leq \prod_{(r,i) \in S} \mathbb{E} \eta_{r,i} = (s/m)^{|S|}.$$

The above is, in a certain sense, a generalization of the CountSketch matrix. For example, we could also select exactly s entries per column to be non-zero, uniformly without replacement.

Now we show the main theorem.

Theorem 2.3.1

As long as $m \simeq \varepsilon^{-2} \log(1/\delta)$ and $s \simeq \varepsilon m$, where m is the row number of matrix Π (just as we have constructed ahead),

$$\forall z : \|z\|_2 = 1, \quad \mathbb{P}_{\Pi}(\|\Pi z\|_2^2 - 1 > \varepsilon) < \delta. \quad (2.6)$$

Proof: Treating σ as an md -dimensional vector and let $E = \|\Pi z\|_2^2 - 1$

To compute the ℓ_2 -norm, we have:

$$\begin{aligned} \|\Pi z\|_2^2 &= z^\top \Pi^\top \Pi z \\ &= \sum_{r=1}^m \langle (\Pi z)_r, (\Pi z)_r \rangle = \sum_{r=1}^m (\Pi z)_r^2 \\ &= \sum_{r=1}^m \left(\sum_{i=1}^d \Pi_{r,i} z_i \right)^2. \end{aligned}$$

Since $\Pi_{r,i} = \eta_{r,i} \sigma_{r,i} / \sqrt{s}$ with $\sigma_{r,i} \in \{\pm 1\}$ and $\eta_{r,i} \in \{0, 1\}$. Then

$$\begin{aligned} \|\Pi z\|_2^2 &= \frac{1}{s} \sum_{r=1}^m \left(\sum_{i=1}^d \eta_{r,i} \sigma_{r,i} z_i \right)^2 \\ &= \frac{1}{s} \sum_{r=1}^m \sum_{i=1}^d \sum_{j=1}^d \eta_{r,i} \eta_{r,j} \sigma_{r,i} \sigma_{r,j} z_i z_j \\ &= \frac{1}{s} \sum_{r=1}^m \sum_{i=1}^d \eta_{r,i}^2 \sigma_{r,i}^2 z_i^2 + \frac{1}{s} \sum_{r=1}^m \sum_{i \neq j} \eta_{r,i} \eta_{r,j} \sigma_{r,i} \sigma_{r,j} z_i z_j. \end{aligned}$$

If $i = j$, using $\eta_{r,i}^2 = \eta_{r,i}$ and $\sigma_{r,i}^2 = 1$, we have

$$\begin{aligned} \frac{1}{s} \sum_{r=1}^m \sum_{i=1}^d \eta_{r,i}^2 \sigma_{r,i}^2 z_i^2 &= \frac{1}{s} \sum_{r=1}^m \sum_{i=1}^d \eta_{r,i} z_i^2 \\ &= \frac{1}{s} \sum_{i=1}^d z_i^2 \sum_{r=1}^m \eta_{r,i}. \end{aligned}$$

Notice that $\sum_{r=1}^m \eta_{r,i} = s$ for every i , and $z \in \mathbb{R}^d$ is a unit vertex, then it equals $\sum_{i=1}^d z_i^2 = 1$. Therefore,

$$E = \|\Pi z\|_2^2 - 1 = \frac{1}{s} \sum_{r=1}^m \sum_{i \neq j} \eta_{r,i} \eta_{r,j} \sigma_{r,i} \sigma_{r,j} z_i z_j \stackrel{\text{def}}{=} \sigma^\top A_{z,\eta} \sigma.$$

Note that $A_{z,\eta}$ is a block diagonal matrix with m blocks, where the r th block is $(1/s)z^{(r)}(z^{(r)})^\top$ but with the **diagonal zeroed out (its diagonal entries is zero)**. Here $z^{(r)}$ is the vector with $(z^{(r)})_i = \eta_{r,i} z_i$

To estimate $\mathbb{P}_\Pi(|\|\Pi z\|_2^2 - 1| > \varepsilon)$, we use Markov's Inequality with p -moment. That is, $\mathbb{P}(|E| > \varepsilon) < \varepsilon^{-p} \cdot \mathbb{E}[|E|^p]$. Thus we need to bound $\mathbb{E}[E^p]$, which is equal to bound $\|E\|_p$. Which technique can we use? Hanson-Wright Inequality!

Noticing that $\mathbb{E}_\sigma[\sigma^\top A_{z,\eta} \sigma] = 0$ since $A_{z,\eta}$ has zero diagonal. To use Hanson-Wright, we need to fix η (so $A_{z,\eta}$ is deterministic), Hanson-Wright gives for all $p \geq 1$ with fixed η ,

$$\|E\|_p = \|\sigma^\top A_{z,\eta} \sigma - \mathbb{E}_\sigma[\sigma^\top A_{z,\eta} \sigma \mid \eta]\|_p \lesssim \sqrt{p} \|A_{z,\eta}\|_F + p \|A_{z,\eta}\|.$$

Taking p -moment **over the randomness of η** , we have

$$\|\|E\|_p\|_p = \mathbb{E}_\eta \left[(\mathbb{E}_\sigma[|E|^p \mid \eta])^{\frac{1}{p} \cdot p} \right]^{\frac{1}{p}} = \mathbb{E}_{\eta,\sigma}[|E|^p]^{\frac{1}{p}} = \|E\|_p$$

and using Minkowski's inequality,

$$\|E\|_p \lesssim \|\sqrt{p} \|A_{z,\eta}\|_F + p \|A_{z,\eta}\|\|_p \lesssim \sqrt{p} \|\|A_{z,\eta}\|_F\|_p + p \|\|A_{z,\eta}\|\|_p.$$

Now we just need to bound $\|\|A_{z,\eta}\|_F\|_p$ and $\|\|A_{z,\eta}\|\|_p$.

Since $A_{z,\eta}$ is block-diagonal, its operator norm is the largest operator norm of any block. The eigenvalue of the r th block is at most $\frac{1}{s} \cdot \|z^{(r)}\|_2^2 \leq \frac{1}{s}$ (see the note below the end of the proof), and thus $\|A_{z,\eta}\| \leq 1/s$ with probability 1. Next, define $Q_{i,j} := \sum_{r=1}^m \eta_{r,i} \eta_{r,j}$, since $\eta_{r,i}^2 \cdot \eta_{r,j}^2 = \eta_{r,i} \cdot \eta_{r,j}$ for η is $\{0, 1\}$ Bernoulli random variable, so that

$$\|A_{z,\eta}\|_F^2 = \frac{1}{s^2} \sum_{i \neq j} z_i^2 z_j^2 \cdot Q_{i,j}.$$

We will show for $p \simeq s^2/m$ that for all i, j , $\|Q_{i,j}\|_p \lesssim p$, where we take the p -norm over η .

Therefore for this p ,

$$\begin{aligned}
\| \|A_{z,\eta}\|_F \|_p &= \mathbb{E}[\|A_{z,\eta}\|_F^p]^{\frac{1}{p}} \\
&= \mathbb{E}[(\|A_{z,\eta}\|_F^2)^{\frac{p}{2}}]^{\frac{1}{p}} \\
&= \| \|A_{z,\eta}\|_F^2 \|_{p/2}^{1/2} \\
&= \left\| \frac{1}{s^2} \sum_{i \neq j} z_i^2 z_j^2 \cdot Q_{i,j} \right\|_{p/2}^{1/2} \\
&\leq \left\| \frac{1}{s^2} \sum_{i \neq j} z_i^2 z_j^2 \cdot Q_{i,j} \right\|_p^{1/2} \quad (\text{non-decreasing property of p-moment}) \\
&\leq \frac{1}{s} \left(\sum_{i \neq j} z_i^2 z_j^2 \cdot \|Q_{i,j}\|_p \right)^{1/2} \quad (\text{Minkowski inequality}) \\
&\leq \frac{1}{s} \cdot \sqrt{p} \cdot \left(\sum_{i \neq j} z_i^2 z_j^2 \right)^{1/2} \\
&\leq \frac{1}{\sqrt{m}}.
\end{aligned}$$

Then by Markov's inequality and the settings of p, s, m ,

$$\mathbb{P}(|\| \Pi x \|_2^2 - 1| > \varepsilon) = \mathbb{P}(|\sigma^\top A_{z,\eta} \sigma| > \varepsilon) < \varepsilon^{-p} \cdot C^p (m^{-p/2} + s^{-p}) < \delta.$$

To show $\|Q_{i,j}\|_p \lesssim p$, we need to use Bernstein's Inequality. Here we introduce its moment form below.

Corollary 2.3.2 Bernstein's Inequality (moment form)

Let $X_1, \dots, X_n$ be independent, each bounded in magnitude by K almost surely. Write $X := \sum_{i=1}^n X_i$ and define $\sigma^2 := \text{Var}[X]$. Then

$$\forall p \geq 1, \quad \|X - \mathbb{E}X\|_p \lesssim \sigma \sqrt{p} + Kp.$$

Suppose $\eta_{a_1,i}, \dots, \eta_{a_s,i}$ are all 1, where $a_1 < a_2 < \dots < a_s$. Now note $Q_{i,j}$ can be written as $\sum_{t=1}^s Y_t$, where Y_t is an indicator random variable for the event that $\eta_{a_t,j} = 1$. The Y_t are not independent, but for any integer $p \geq 1$ their p th moment is upper bounded by the case that the Y_t are independent Bernoulli each of expectation s/m (this can be seen by simply expanding $(\sum_t Y_t)^p$ and comparing with the independent Bernoulli case monomial by monomial in the expansion). Thus Bernstein applies, and as desired we have

$$\|Q_{i,j}\|_p = \left\| \sum_t Y_t \right\|_p \lesssim \sqrt{s^2/m} \cdot \sqrt{p} + p \simeq p.$$

Thus we end the whole proof. ☺

Note:-

Why $\|A_{\eta,z}\| \leq 1$? Here we compute it: noticing that

$$\begin{aligned}\|A_{\eta,z}\| &= \sup_{\|x\|_2=1} \|A_{\eta,z}x\|_2 \\ &= \sup_{\|x\|_2=1} \|z^{(r)}(z^{(r)})^\top x\|_2 \\ &= \sup_{\|x\|_2=1} \|k \cdot z^{(r)}\|_2\end{aligned}$$

Where $k = (z^{(r)})^\top x$. From the Cauchy Inequality, it's not hard to know that $k \leq 1$, and we also know $\|z^{(r)}\| \leq 1$, so $\|A_{\eta,z}\| \leq 1$.

2.3.2 Fast JL Transforms

Another approach for obtaining fast JL was investigated by Ailon and Chazelle. This approach gives a running time to compute Πx of roughly $O(d \log d)$, which is faster than the sparse JL approach when x is sufficiently dense. They called their transformation the *Fast Johnson–Lindenstrauss Transform (FJLT)*. A construction similar to theirs, which we will analyze here, is the $m \times d$ matrix Π defined as

$$\Pi = \frac{1}{\sqrt{m}} SHD \tag{5.9}$$

where the components are:

- $S \in \mathbb{R}^{m \times n}$: a **sampling matrix with replacement**. Each row has a single 1 in a uniformly random position and 0 elsewhere; the rows are independent.
- $H \in \mathbb{R}^{n \times n}$: an *unnormalized bounded orthonormal system*, which means H satisfying

$$H^\top H = d \cdot I, \quad \max_{i,j} |H_{i,j}| \leq 1.$$

Typical example is Hadamard matrix.

Note:-

For $d = 2^k$, the $d \times d$ unnormalized Hadamard matrix (with 0-based indexing) has entries

$$H_{i,j} = (-1)^{\langle \vec{i}, \vec{j} \rangle \bmod 2}, \quad 0 \leq i, j < d,$$

where $\vec{i}$ is the k -dimensional binary vector obtained by writing i in binary.

- $D = \text{diag}(\alpha)$: a diagonal matrix where α is a vector of n independent Rademacher random variables.

Chapter 3

Linear Algebra Applications

3.1 Approximate matrix multiplication

3.2 Subspace embeddings

3.2.1 Oblivious subspace embeddings

3.3 least square regression

3.4 Approximate low-rank approximation

3.5 Projection-cost preserving sketches